VTool-R1: VLMs Learn to Think with Images via Reinforcement Learning on Multimodal Tool Use

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Outline

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Why Do VLMs Still "Think" Only in Text?

Background:

- RL fine-tuning (RFT) gave LLMs long CoT, self-correction and **learned tool use** (Search-R1).
- Recent VLM "R1" works (R1-V, Vision-R1, R1-VL, ...) copy this recipe.
- **But their reasoning stays text-driven:** the image is encoded once; every later step is text.

Why it matters

- A *six-fingered* hand: "how many fingers?" $\rightarrow$ the text prior "five" can beat the pixels.
- Inference-time visual CoT (Visual Sketchpad, ReFocus) helps, but only with GPT-4o-class models; small open models fail.

The Core Question

Can we **train** a small open VLM, with RL and only an outcome reward, to produce *intermediate visual steps* (edited images) and reason over them, i.e. to "think with images"?

The Missing Piece

No prior RFT framework puts image-editing tool calls *inside* the rollout, so the model never learns *when* or *how* to look again.

Three Roadblocks to Multimodal Chains of Thought

1. **Language shortcuts**
   Text-only CoT lets the decoder drift to a plausible-sounding answer that the image contradicts; nothing forces the model to re-inspect visual evidence.

2. **Prompting does not transfer to small models**
   Prompted tool use works for GPT-4o, but open Qwen2.5-VL 3B/7B fail to follow tool-use instructions: prompting them to use tools *lowers* accuracy (3B chart: 51.8 $\rightarrow$ 24.6).

3. **Training with tools in the loop is awkward**
   Generation must pause for external Python execution, the edited image must be fed back as a new input, and rewards on the tool calls themselves invite reward hacking.

Proposed Solution: VTool-R1

Contribution 1: RFT Framework with Visual Tools

- Python image-editing tools are executed *during* RL rollouts.
- Two-pass rollout: reason $\rightarrow$ edit image $\rightarrow$ reason over original $\oplus$ edited image.
- Built on GRPO; trained with an **outcome-only** 0/1 reward.

Contribution 2: First Learned "Thinking with Images"

- Claimed first work to make VLMs *learn* to interleave visual steps with text CoT.
- Model decides *whether* a tool helps; it is never rewarded for using one.
- Validated on chart and table QA (ReFocus splits), Qwen2.5-VL 3B/7B/32B.

Goal: turn "visual editing as chain of thought" from a GPT-4o-only prompting trick into a skill that small open VLMs acquire through RL.

Thinking with Images: What a Trained Rollout Looks Like

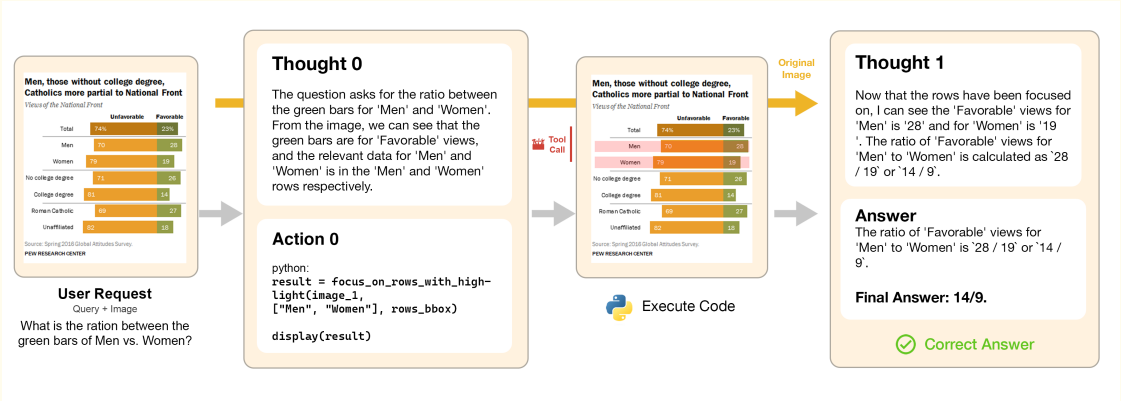
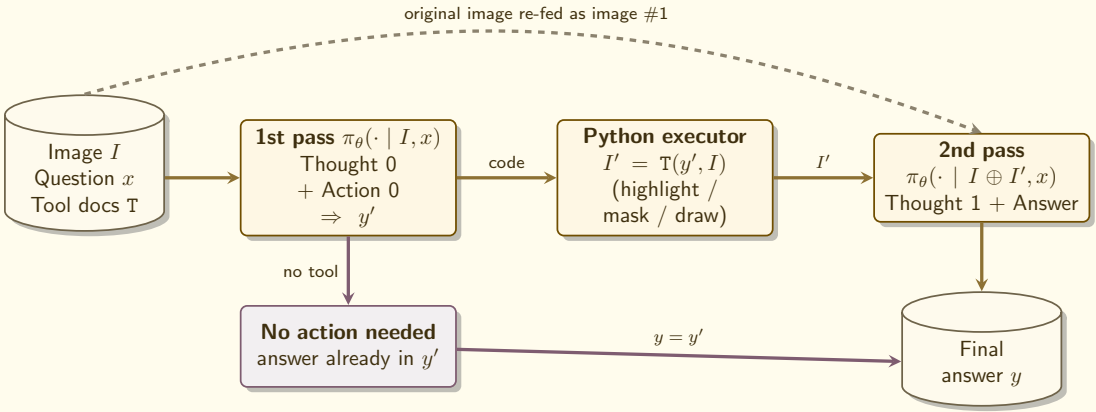


Figure: After RFT, the 3B model reasons in *Thought 0*, emits a tool call that highlights the "Men" and "Women" rows, receives the edited chart *alongside* the original, and reads the "Favorable" values (28 vs. 19) in *Thought 1*. The visual step narrows attention to exactly the evidence the question needs.

Inference & Rollout with a Visual Chain of Thought



$$y' \sim \pi_\theta(\cdot \mid I, x), \qquad I' = \mathtt{T}(y', I)$$

$$y \sim \pi_\theta(\cdot \mid I, x; \mathtt{T}) = \pi_\theta\big(\cdot \mid I \oplus \mathtt{T}(y', I),\, x\big)$$

- $\oplus$: original and edited image given as **two image inputs** (needs a multi-image VLM).
- At most **one** tool turn; several tool calls may be chained inside it.

Prompt Format and the Visual Editing Toolset T

Structured output format

```
THOUGHT 0:  rows ‘‘Men’’ and ‘‘Women’’ hold
the answer
ACTION 0:
  r = focus_on_rows_with_highlight(
      image_1, ["Men","Women"], rows_bbox)
  display(r)
OBSERVATION: <edited image = image #2>
THOUGHT 1:  Men = 28, Women = 19...
ANSWER: ... FINAL ANSWER: 14/9
TERMINATE
```

Prompt includes tool signatures, 9 requirements (e.g. “only one ACTION”), few-shot examples, and the **bounding boxes** of rows/columns (or chart bars).

Tools (Python + OpenCV, from ReFocus)

Tool	Applies to	Effect
Highlight	columns / rows	semi-transparent red overlay
Mask	columns / rows	white mask over irrelevant parts
Draw	columns / rows	solid red bounding box
Chart variants	bars (by x/y value)	highlight / mask bars

Intuition

Tools simulate **selective visual attention**: the model edits the image to isolate the evidence, then reads it again, as a human would with a highlighter.

Multi-Modal GRPO with Tool Use: Training Pipeline

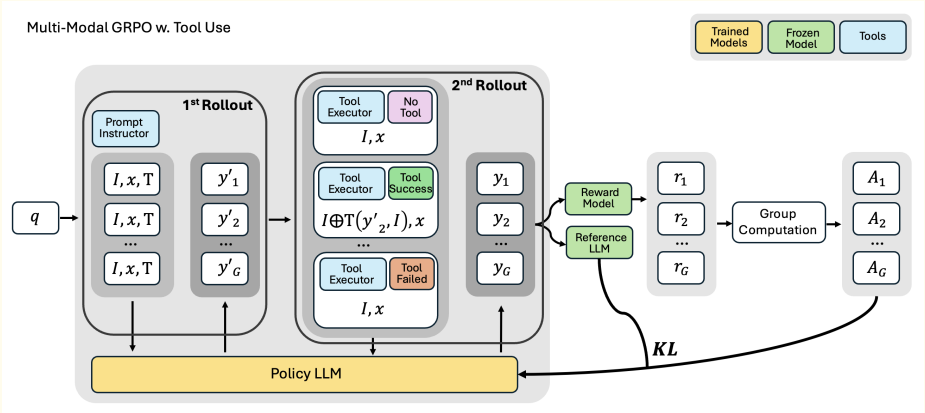


Figure: For each query q, G first-pass responses y'_i go through the tool executor: *no tool* (answer from I, x), *tool success* (2nd pass on $I \oplus \mathtt{T}(y'_i, I)$), or *tool failed* (fall back to I, x). Only the final y_i are scored, turned into group-relative advantages A_i, and used to update the policy under a KL penalty to the frozen reference.

Core Design Choices and Why

1. Optimize only the final response y

- The intermediate tool-call text y' is not directly optimized.
- *Why:* the goal is final answer quality; the model should decide on its own *whether* a tool helps. More end-to-end.

2. Edited image fed back as a 2nd input

- Unlike Search-R1, the tool output is not spliced into the token sequence.
- *Why:* the VLM must jointly attend to I and I'; requires multi-image capability.

3. Single tool turn

- One ACTION, possibly with several chained edits.
- *Why:* multi-turn editing needs stronger multi-image VLMs; left as future work.

4. GRPO, no format reward

- Critic-free group baseline: cheaper and more stable than PPO for 16k-token, multi-image rollouts.
- Structured template + few-shot examples already give reliable formatting.

Central insight

Tools are only *available* in the rollout and only correctness is rewarded, so tool use survives only where it actually helps.

Training Objective

RL objective with tools in the rollout (KL-regularized, Bradley–Terry reward r_ϕ):

$$\max_{\pi_\theta} \; \mathbb{E}_{[I,x]\sim\mathcal{D},\; y\sim\pi_\theta(\cdot|I,x;\mathtt{T})}\big[r_\phi(I,x,y)\big] \;-\; \beta\,\mathbb{D}_{\mathrm{KL}}\big[\pi_\theta(\cdot \mid I,x;\mathtt{T}) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid I,x;\mathtt{T})\big]$$

GRPO surrogate, with a group $\{y_i\}_{i=1}^{G} \sim \pi_{\mathsf{old}}(\cdot \mid I,x;\mathtt{T})$:

$$\mathcal{J}_{\mathsf{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|y_i|}\sum_{t=1}^{|y_i|}\min\Big(\rho_{i,t}(\theta)\hat{A}_{i,t},\; \mathrm{clip}\big(\rho_{i,t}(\theta),1-\epsilon,1+\epsilon\big)\hat{A}_{i,t}\Big) - \beta\,\mathbb{D}_{\mathrm{KL}}\big[\pi_\theta \,\|\, \pi_{\mathrm{ref}}\big]\right]$$

$$\hat{A}_{i,t} = \frac{r_i - \mathrm{mean}(\mathbf{r})}{\mathrm{std}(\mathbf{r})}$$

- $\rho_{i,t}$: token-level probability ratio $\pi_\theta/\pi_{\mathsf{old}}$ (written $r_{i,t}$ in the paper).
- $\hat{A}_{i,t}$: same advantage for every token of y_i, no critic.
- Only tokens of the final y_i get gradient; I' comes from the environment.

Reward Design: Outcome Only, and Why Process Rewards Fail

Reward: $r_i = 1$ if a lightweight **LLM judge** says the predicted answer matches the ground truth, else 0 ("pseudo rule-based"; string match is too brittle for free-form chart/table answers).

Outcome reward (used)

0/1 on final-answer correctness only.

Tool use rises and falls with its usefulness; accuracy improves steadily.

Penalize failed tool calls

Model learns the safest policy:

tool usage collapses to zero.

Bonus for successful tool use

Model emits calls that merely *trigger* the success check;

hacking persists even with a stricter verifier.

Takeaway

Rewarding the *process* of tool use is easy to game. Rewarding only the *outcome* is the most robust signal found by the authors (evidence is reported qualitatively, without a table).

Experimental Setup

Data (exact ReFocus splits)

- **Table split:** VWTQ (750 QA, Wikipedia tables), VWTQ_syn (250, synthetic styles), VTabFact (250, entailment). 70/30 train/test, no validation set.
- **Chart split:** ChartQA; 14,344 train / 813 val; test = 444 horizontal + 382 vertical bar charts.

Models & baselines

- Qwen2.5-VL 3B / 7B / 32B: *Pure Run* (no tools) and *Tool Use* (prompted, no RFT; regenerate on failed call).
- R1-VL 2B/7B (general visual RL), DeepEyes 7B (concurrent), GPT-4o (upper bound).

Training (VeRL)

Optimizer	AdamW, lr 10^{-6}, wd 10^{-2}
Max sequence	16,384 tokens
Global batch	32 (table) / 256 (chart)
Group size G	5
KL coeff. β	10^{-2}
Decoding	temperature 1.0, bf16
Hardware	3B/7B: 8/16×H100; 32B: 8×H200

Metric: answer accuracy (%) judged against ground truth.

Main Results: Does RFT Teach Useful Tool Use?

Accuracy (%) of Qwen2.5-VL before and after VTool-R1

Method	3B		7B		32B	
	Chart	Table	Chart	Table	Chart	Table
Pure Run (no tools)	51.8	41.3	76.2	64.7	**88.0**	**86.2**
Tool Use (prompted)	24.6	24.3	53.4	41.1	85.0	76.0
VTool-R1 (RFT)	**64.0**	**57.9**	**80.7**	**71.7**	86.7	84.5

Key Takeaways:

- **+12.2 / +16.6** points over Pure Run at 3B (chart / table); **+4.5 / +7.0** at 7B.
- Prompting small models to use tools **costs 20–27 points**; RFT turns the same tools from a liability into a gain (+39.4 / +33.6 at 3B).
- **At 32B the gain disappears:** VTool-R1 recovers most of the tool-use drop but stays 1.3 / 1.7 points below Pure Run.

Comparison with Other RL-Tuned VLMs

Model	Size	Chart	Table
R1-VL (general visual RL)	2B	10.4	8.6
R1-VL (general visual RL)	7B	63.8	45.4
DeepEyes (RL + zoom tools)	7B	60.0	–
VTool-R1	3B	64.0	57.9
VTool-R1	**7B**	**80.7**	**71.7**
GPT-4o, Pure Run (upper bound)	–	82.9	75.7
GPT-4o, Tool Use (ReFocus)	–	80.5	77.0

- VTool-R1 3B already beats R1-VL 7B; the 7B model is within **2.2 / 4.0** points of GPT-4o.
- vs. concurrent DeepEyes: **80.7 vs. 60.0** on charts, attributed to tools and tasks matched to structured images.
- Caveat: R1-VL and DeepEyes were not trained on these splits, so part of the gap is in-domain training, not tool use alone.

Training Dynamics: Tool Use Is Not Monotonic

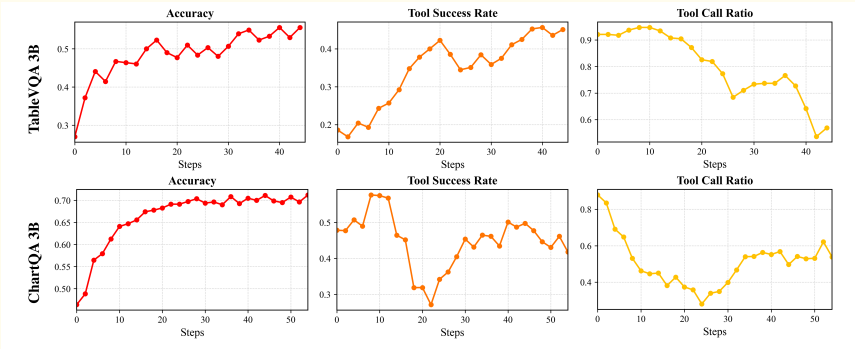
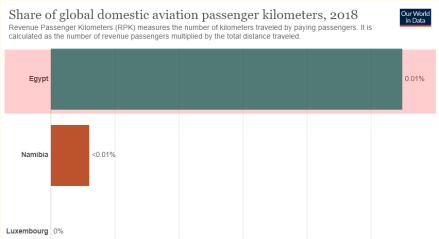


Figure: 3B training curves, TableVQA (top) and ChartQA (bottom). Accuracy rises (table ≈0.27→0.55, chart ≈0.46→0.71) and saturates in ∼50 steps, while the tool-call ratio *falls* (table 0.92→0.57; chart dips to 0.28, recovers to ∼0.55).

- Prompt-driven over-use early on; later the model is **selective** and skips tools when unnecessary.
- Tool success (proxy: code runs, returns an image) climbs on tables, fluctuates on charts; 32B keeps a higher call ratio.

Failure Analysis



Case (a). Q: "What is the value of the largest bar?" The tool correctly highlights Egypt and the 1st pass reads **0.01**, but the 2nd pass on the edited image says ≈0.001 and outputs "Egypt". GT: 0.01.

Four failure categories

(a) **Right visual step, wrong inference:** the 2nd rollout misreads or overrides a correct first-pass reading.

(b) **Flawed edit:** the drawn box occludes the numbers it was meant to emphasize.

(c) **Wrongly skips tools:** decides "no action needed" and answers incorrectly.

(d) **Invalid code:** hallucinated APIs (e.g. `get_children()`) fail, so no visual step is produced.

Observation

Since y' gets no gradient, a correct first-pass reading can be lost in the second pass; the visual step helps only if the 2nd pass actually uses it.

Current Limitations

Stated by the authors

- **Single tool turn only**: no iterative edit–reason loops.
- **Narrow scope**: selective attention on charts and tables with a fixed toolset.
- **No oracle for tool correctness**: "success" only means the code ran.

Our observations

- **Tools need given bounding boxes**: row/column/bar boxes are supplied in the prompt, which simplifies grounding.
- **No benefit at 32B**: gains shrink with scale; Pure Run wins at 32B.
- **Missing ablation**: no RFT run *without* tools, so the gain from RL alone vs. RL + tools is not isolated.
- **Reward-design evidence is qualitative**; the LLM judge is itself a possible noise source.

Conclusion & Takeaways

1. **Framework:** VTool-R1 puts Python image-editing tools inside GRPO rollouts, so a VLM reasons over the original image $\oplus$ its own edited image.

2. **Outcome-only reward is enough:** the model learns *when* to use tools without process supervision, while process rewards get hacked.

3. **Results:** 3B/7B Qwen2.5-VL gain up to **+16.6** points over direct answering; 7B approaches GPT-4o on chart and table QA.

4. **Outlook:** multi-turn editing, richer (generative) tools, and broader tasks beyond structured images.

Thank You!

Questions?

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